Huawei Access Node BAS-ACC-28 Optical Transport Failure SOP (ALARM_7617)

Node: BAS-ACC-28 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7617

Description

This Standard Operating Procedure (SOP) addresses the critical optical transport failure on Huawei Access Node BAS-ACC-28 at Basavanagudi POP, triggered by the ALARM_7617 alarm code. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss in the optical transport network. This is due to the inherent nature of optical signals, which are susceptible to degradation caused by physical stress, environmental factors, and hardware malfunctions. In the context of Huawei's Access Node, the optical transport failure is precipitated by the SFP/XFP's inability to maintain optimal optical power levels, resulting in a degradation of signal quality. This is further exacerbated by the physical properties of fiber optic cables, which are prone to cuts and bends, leading to signal attenuation and loss. The Huawei Access Node's role in the access network necessitates a robust and reliable optical transport infrastructure, which, when compromised, can lead to critical service outages. Therefore, it is essential to promptly identify and rectify the root cause of the optical transport failure to prevent extended service disruptions.

Pre-requisite Checks

- item: Ensure remote hands are dispatched to Basavanagudi POP to facilitate physical access to the node., description: Verify that the remote hands are equipped with the necessary tools and expertise to perform the required maintenance tasks.

- item: Verify the status of redundant links to determine if the failure is isolated to a single link or affects the entire network., description: This will help in determining the scope of the failure and the necessary corrective actions.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to physical stress or environmental factors.	High
Fiber cuts or bends causing signal attenuation and loss.	High
Rx/Tx power loss due to hardware malfunction or misconfiguration.	Medium

Diagnostic Commands & Expected Outputs

```
Command: display optical-fiber status, Expected Output: Optical fiber status: normal, Description: This command will display the current status of the optical fibers, including any errors or alarms. The output will help in identifying potential issues with the fiber itself or the SFP/XFP.
```

```
Command: display sfp-xfp status, Expected Output: SFP/XFP status:
normal, Description: This command will display the current status of
the SFP/XFP, including any errors or alarms. The output will help in
identifying potential issues with the SFP/XFP or the fiber optic
cable.

Command: display rx-tx status, Expected Output: Rx/Tx status: normal,
Description: This command will display the current status of the
receive and transmit paths, including any errors or alarms. The output
will help in identifying potential issues with the Rx/Tx power levels
or hardware misconfiguration.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module with a new one., Command: replace sfp-xfp slot 1/1/1 with new sfp-xfp, Description: This step will replace the faulty SFP/XFP module with a new one, which should resolve the issue if the failure was due to SFP/XFP degradation. Ensure that the new SFP/XFP is properly seated and secured before proceeding.

Step 2: Step: Verify the status of the fiber optic cable and replace it if necessary., Command: display optical-fiber status, Description: This step will verify the status of the fiber optic cable and replace it if necessary. Ensure that the new fiber optic cable is properly connected to the SFP/XFP before proceeding.

Step 3: Step: Adjust the Rx/Tx power levels to optimal values., Command: adjust rx-tx power levels to optimal values, Description: This step will adjust the Rx/Tx power levels to optimal values, which should resolve the issue if the failure was due to Rx/Tx power loss. Ensure that the power levels are properly calibrated before proceeding.

Post-Check Verification

```
Command: display optical-fiber status, Expected Output: Optical fiber
status: normal, Description: This command will verify that the optical
fiber is functioning correctly and that the failure has been resolved.

Command: display sfp-xfp status, Expected Output: SFP/XFP status:
normal, Description: This command will verify that the SFP/XFP is
functioning correctly and that the failure has been resolved.

Command: display rx-tx status, Expected Output: Rx/Tx status: normal,
Description: This command will verify that the receive and transmit
paths are functioning correctly and that the failure has been
resolved.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module., Command: restore sfp-xfp slot 1/1/1 to original, Description: This step will restore the original SFP/XFP module if the replacement procedure was unsuccessful. Ensure that the original SFP/XFP is properly seated and secured before proceeding.

- Step: Re-adjust the Rx/Tx power levels to their original values., Command: adjust rx-tx power levels to original values, Description: This step will re-adjust the Rx/Tx power levels to their original values if the adjustment procedure was unsuccessful. Ensure that the power levels are properly calibrated before proceeding.